Food Loss and Waste Analysis

Where Reported Loss Concentrates: Pareto Analysis

PROJECT REPORT

Project Charter

Project goal: To analyse food supply stages and commodities with the recorded food loss in the database (2000 - 2024), and identify where the loss is concentrated along the food supply stages.

Project Overview: This project analyses the FAO Food Loss and Waste Database (2000 - 2024) to identify where reported food loss concentrates along the supply chain. It uses 21,777 records from 115 countries across 19 food supply stages and the findings are presented in an interactive Tableau Public dashboard. Pareto analysis of high-loss records (reported loss of 20% or more) identifies the food supply stages.

Project Scope:

Item	Scope
Data Source	FAO Food Loss and Waste Database, Technical Platform on the Measurement and Reduction of Food Loss and Waste. https://www.fao.org/platform-food-loss-waste/flw-data/en/
Period	2000-2024
Records	21,777 records after cleaning and filtering
Measure	Median of reported loss_percentage
Analysis	Charts, Pareto of high-loss records (20% or more)
Deliverables	Tableau Public dashboard, report, README

Data Cleaning log

Step	Rows Left
Raw Download	31,455
Remove exact duplicates on all 18 columns	31,275
Remove rows with a blank supply chain stage	27,884

- 9 of 18 columns retained and 9 dropped (region, loss_quantity, activity, treatment, cause_of_loss, sample_size, reference, url, notes).
- cpc_code lost its leading zeros on saving; it is not used in the analysis.
- loss_percentage_original mixes two scales; it is not used in the analysis.
- Country names are used as published in the dataset.

The dashboard excludes them with these Tableau filters:

- **Food Supply Stage:** exclude "Whole supply chain"
- **Country:** exclude World and the 13 regional aggregate names (Africa, Asia, Europe, Central Asia, Northern Africa, Northern America, South-Eastern Asia, Southern Asia, Sub-Saharan Africa, Western Africa, Western Asia, Latin America and the Caribbean, Australia and New Zealand).

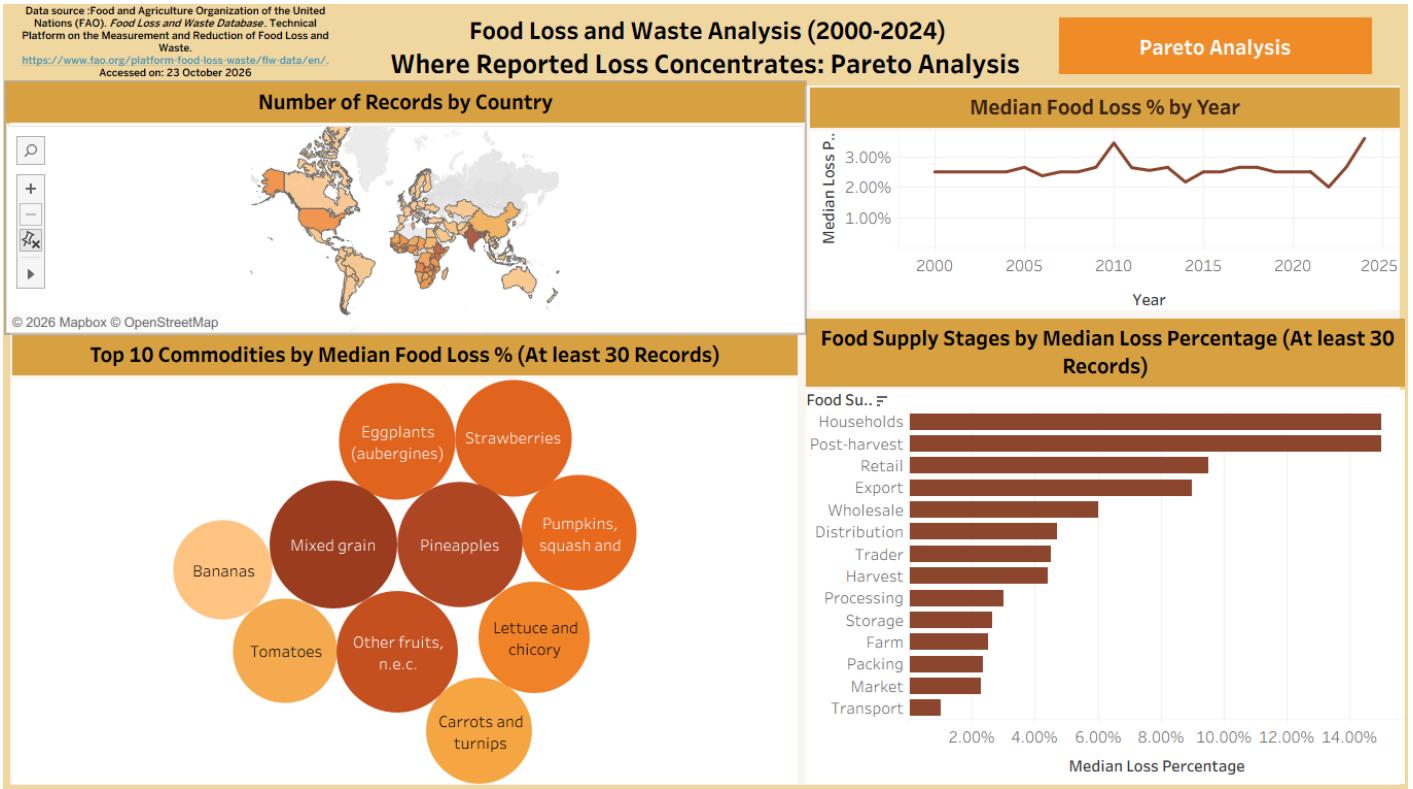
Filter	Rows Left
Cleaned File	27,884
After excluding Whole supply chain, World, and the 13 regional aggregate names.	21,777

Methodology

1. **Data Collection:** Downloaded the FAO Food Loss and Waste Database, covering reported food loss from 2000-2024. (The dataset was only downloaded for years 2000-2024).
2. **Data Cleaning:** Removed exact duplicate rows and rows with missing food supply-chain stages, reducing the dataset from 31,455 raw rows to 27,884 rows. Retained nine columns out of the original 18 columns for analysis.

3. **Data Filtering in Tableau:** Excluded whole-supply-chain records and World/regional aggregates to avoid double-counting errors and improve comparability. The final analysis dataset contained 21,777 records from 115 countries across 19 supply-chain stages.
4. **Data Analysis:** Examined reported food-loss patterns by country, year, commodity, and supply-chain stage. Used the number of records for country-level data coverage.
5. **Statistical Analysis:** Used median reported loss percentages to compare years, commodities, and supply-chain stages because the loss-percentage data are skewed. Applied a minimum of 30 records for the commodity and supply-chain-stage comparisons where specified.
6. **Pareto Analysis:** Classified records with reported food loss of 20% or more as high-loss records, using a project-defined threshold. Counted these records by supply-chain stage and calculated cumulative percentages to identify the stages accounting for the largest share of high-loss records.
7. **Dashboard:** Built an interactive Tableau Public dashboard to present the number of records by country, median loss % by year, top 10 commodities by median loss % (at least 30 records), food supply stage by median loss % (at least 30 records), and Pareto analysis.
8. **Validation and Interpretation:** Checked reported findings against the cleaned data and documented limitations, including uneven record coverage and unweighted medians.

Dashboard- Food Loss and Waste Analysis



Metric Explanation

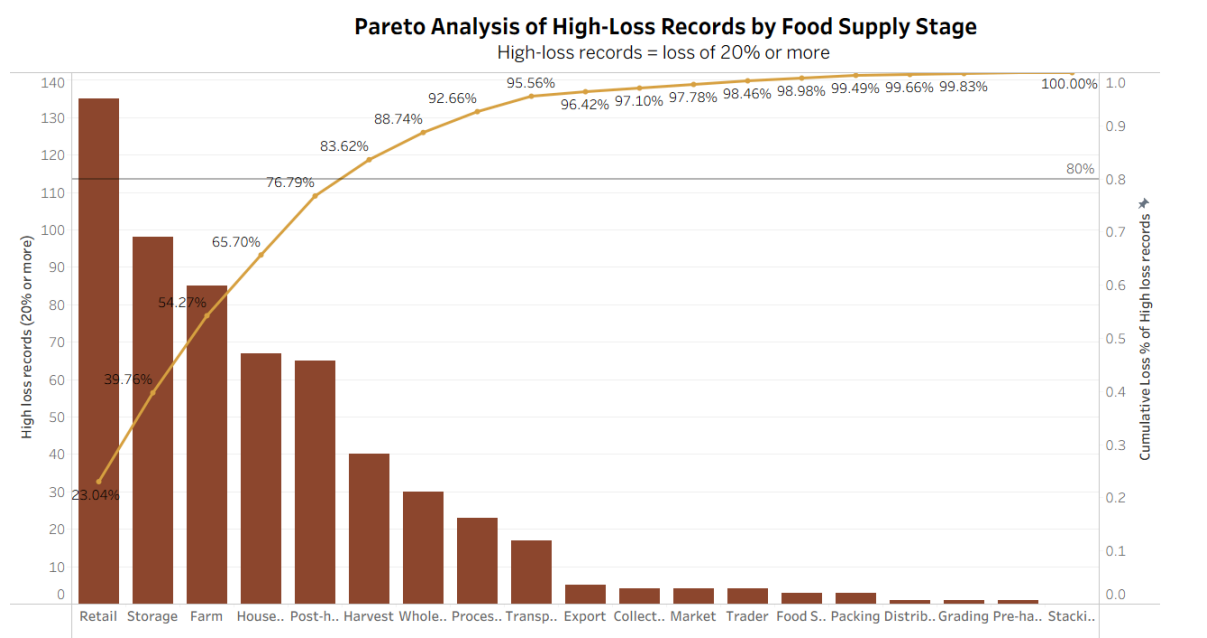
Metric	Explanation
Median loss %	Median of loss_percentage
Number of records	Count of loss_percentage
High loss record	A record with reported loss of 20% or more. *This is a project-defined threshold
Cumulative % of high-loss records	Running total of high-loss records divided by the total number of high-loss records (586), with stages in descending order.

Key Insights

- The median reported loss is 2.5% across 21,777 records.
- Post-harvest and household stages stand out. Both reach a 15.00% median, followed by Retail at 9.50%. Transport records the lowest loss at 1.00%.
- Among commodities with at least 30 records, Mixed grain (12.00%), Pineapples (11.52%), and Other fruits, n.e.c. (10.85%) rank highest.
- The number of records by country is uneven. India (1,233 records), Ethiopia (992), and Kenya (909) dominate, which limits country-level comparison.

Pareto Analysis

A record is classified as high-loss when its reported loss is 20% or more. This is a project-defined threshold, used only for the Pareto analysis.



Calculation Method: High-loss records were grouped by food supply-chain stage and counted. Each stage's percentage contribution was calculated by dividing its record count by the total of 586 high-loss records and multiplying by 100. The cumulative percentage was then calculated by adding each stage's contribution in descending order of record count. The stages were ranked from highest to lowest record count to identify the point at which the cumulative share exceeded 80%.

Key Insights

- High-loss records are concentrated. Out of 586 high-loss records, six stages hold 83.62%: Retail, Storage, Farm, Households, Post-harvest, and Harvest.
- **Retail is the largest contributor.** It accounts for 135 records (23.04%).
- **The 80% line is crossed at the sixth stage.** The first five stages cover 76.79%, and adding Harvest makes the total 83.62%.

Limitations

- Medians are not weighted by production volume, so they show typical reported loss, not total food lost.
- Some countries, commodities, and stages have few records, so their medians are less stable, and years 2022 to 2024 also have fewer records.
- The dataset only provides 'loss_percentage' for all stages.
- Whole-supply-chain, World, and regional aggregate rows are excluded to avoid double counting.
- A record is classified as high-loss when its reported loss is 20% or more. This is a project-defined threshold, used only for Pareto analysis.

Sources

- FAO. Food Loss and Waste Database. Technical Platform on the Measurement and Reduction of Food Loss and Waste. fao.org/platform-food-loss-waste/flw-data. (Accessed on: 23 October 2026)

Dashboard:

https://public.tableau.com/views/FoodLossandWasteAnalysis/FoodLossandWasteAnalysis

Pareto Analysis:

https://public.tableau.com/views/FoodLossandWasteAnalysis/ParetoAnalysisofHigh-LossRecordsbyFoodSupplyStage

Made by: Navita Arora